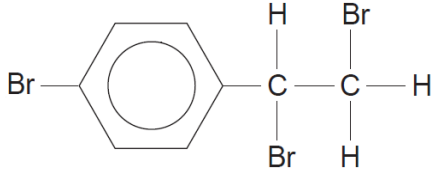


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)			Indicative content mass of carbon in 5.35 g of compound B is 1.50 g molecular formula is C₈H₇Br₃ <i>M_r</i> is 343 there is a chiral carbon atom formula of silver bromide is AgBr mass of bromine in 4.77 g of silver bromide is 2.03 g NaOH hydrolyses aliphatic C—Br bonds % Br in hydrolysis product is 46.6 % Br in compound B is $\frac{3 \times 79.9 \times 100}{343} = 69.9$ therefore two of the three bromine atoms in compound B are removed ⇒ they must be in the side chain bromine atom not removed must be on the ring compound B is  accept aromatic bromine in any position		3	3	6	2	2